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(Key and Solutions for AIMCAT2415)

Key

SECTION – I

1. B
2. C
3. D
4. B
5. B
6. D
7. B
8. C
9. D
10. B
11. C
12. B
13. D
14. D
15. B
16. B
17. C
18. 1423
19. D
20. A
21. C
22. 4213
23. D
24. 3421

SECTION – II

1. 170
2. C
3. 340
4. B
5. B
6. 59
7. D
8. C
9. 244
10. D
11. A
12. C
13. C
14. D
15. A
16. C
17. 6
18. 3
19. B
20. D

SECTION – III

1. 4
2. –36
3. 10
4. B
5. C
6. A
7. C
8. D
9. B
10. 6
11. 57
12. A
13. C
14. B
15. D
16. 6720
17. C
18. A
19. 16
20. 4860
21. D
22. A

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 509

1. Considering the lines "...Plato not only identified various specific weaknesses in human cognition, he also offered powerful proposals for how to overcome these biases and improve our reasoning and behaviour", shows that Plato not only analysed core human biases, as given in detail in the 4th para of the passage, he also proposed solutions on how to overcome these biases. The question talks about proposals that Plato will not endorse, keeping in mind the biases mentioned in the passage.
Option A: This option proposes interacting with people having diverse opinions to better understand an issue. This will directly counter the confirmation bias, where people believe only what they want to believe, which means that they have only their point of view to look at. However, to overcome this, interacting with different people from diverse backgrounds and having diverse opinions would be a great idea. Option A is not the answer.
Option B: This cannot be a way to overcome the biases that Plato mentioned because doing this would show confirmation bias, where a person only believes what he/she wants to believe and takes steps to reinforce their beliefs rather than looking at other perspectives. Plato will not endorse this proposal to overcome biases. Hence, Option B is the answer.
Option C: This talks about reframing a problem to give it a fresh view which would affect a change in the way the problem is viewed, which in turn might help find a solution to the problem. This is a proposal that would counter the framing bias (reversing their opinions about identical propositions based on the language in which the propositions are presented) mentioned in the passage. Plato will endorse this proposal and so it is not the answer.

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Option D: This talks about taking time to come up with options and the considering and weighing the options before making assertions. This proposal would directly contradict the availability bias which is all about asserting what first comes to mind and Plato would recommend this as a viable solution to the bias. Hence, option D is not the answer.
Choice (B)

2. Elusive means something that cannot be easily found. According to Socrates, philosophical enquiry is elusive or difficult to achieve because truth is not always apparent or something that can be found at the surface. It can be found only by delving deeper into thoughts, while keeping in mind the biases we ourselves have that could keep us from finding this truth. People tend to take things at face value and trust their first impressions.
Option A: This talks about people being misled by their first impressions about the world and other people, which can be directly understood from the line "...simply trusting our first impressions and beliefs is a recipe for misapprehending the world.". Hence, option A is not correct.
Option B: This option is another way of saying that people are prone to go with first impressions and don't make efforts to understand the root of the issue. This can again be inferred from the last para of the passage and so option B is not the answer.
Option C: This talks about people being swayed by different perspectives. However, this is not something Socrates mentions when he calls philosophical truth "elusive". This option cannot be inferred from the passage and so option C is the correct answer.
Option D: Socrates mentions the reason why philosophical enquiry is elusive and can be picked up from the line "...and becoming acutely aware of our own susceptibility to certain forms of deception, are we likely to get closer to the truth". Getting rid of these would bring people closer to the truth. Option D is not the right answer.
Choice (C)

3. The habits and processes that lead people to false conclusions, as mentioned in the passage, are confirmation bias, availability bias, framing, a cognitive version of loss aversion, representativeness heuristic, and a version of anchoring. The examples given in the option must suit one of these habits/processes.
Option A: This is an example of confirmation bias as gathered from the line "...people believing what they want or what they are predisposed to believe (confirmation bias)". He is predisposed to like this candidate and chooses to ignore any negative news that might oppose his point of view. Hence, A is not the answer.
Option B: This is also another example of confirmation bias, where, Stephen, who is racist by nature, believes what he wants to believe, even though there is contradicting evidence against his opinion. Option B is not the answer.
Option C: It is important to understand that millions of people buy lottery tickets and the odds of winning are infinitesimally small. Hugh made his decision to buy the tickets based on the sample size of one person and did not consider the millions who didn't win. This is an example of representativeness heuristic, as can be gathered from the part "...making false inferences based on the size and representativeness of a sample of a broader population". Hence, option C is not the correct answer.
Option D: This is an example of herd behaviour and it is not an example of any of the habits and processes that lead people to false conclusions that Plato mentions in the passage. The closest that it can be is the representativeness heuristic, but that cannot be the case here because Audrey does not infer that the medicine will work because of a small sample. Instead, she takes the medicine even though she knows that it does not work. Option D is the answer.
Choice (D)

4. In the passage, the availability bias that the author is talking about is that people tend to remember what comes first to mind, in terms of recency. The example that he uses to endorse this is that Russell's articulation of the concept of availability bias as understood from the lines "...Because every age...they differed little from our own."" came in 1950, way before Kahneman coined the catchphrase "availability bias". Because of this, everyone remembers Kahneman's concepts and catchphrase and forget that it was already previously conceptualized by Russell.
Option A: This is not true and can't be understood from the passage. The passage only talks about Russell's articulation of availability bias and not other core human biases. Option A is not the answer since it is not the example that the author mentions.
Option B: This option talks about Kahneman's articulation of availability bias being more popular than Russell's. This is the exact example that the author uses to demonstrate the concept of availability bias. Option B is the correct answer.
Option C: This says that Kahneman coined the catchphrase after reading Russell's articulation of the same, which is not mentioned at all in the passage and is out of scope. Option C is not the answer.
Option D: This says that Kahneman's use of catchwords made his ideas more popular than Russell's, indicates that this was deliberately done by Kahneman. This is not true. It just so happens that people tend to remember contemporary articulation of an idea because of the catchwords used, rather than any historic references of the same idea. Option D is incorrect.
Choice (B)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 472

5. Option A: It is not the mismatch in tolerance of the μ-opioid receptors that the author calls fatal. It is the mismatch in tolerance caused by the difference in class/concentration of those receptors. This option misrepresents the information. Option A is not the answer.
Option B: The fatality of opioids is because different parts of the body can take different doses or are tolerant to different doses because of the variation in class and concentration of μ-opioid receptors. So, that makes them fatal as one part cannot take what another part can. Option B is the answer.
Option C: The author doesn't really compare the number of μ-opioid receptors with those of other substances to drive home the point of their fatality. Also, the option seems to suggest that the fatality is because of increased tolerance until a dose is dangerous. The fatality is because of mismatch of tolerance as the author attributes to the deadliness of opioids. Option C is not the answer.
Option D: The author hasn't discussed variation in receptor response. The closest he comes to is mentioning that there are different classes and concentrations of receptors which causes tolerance mismatch. Option D is not the answer.
Choice (B)

6. According to the passage cocaine is fatal when one loses tolerance and then takes the same dose. Opioids too are fatal for the same reason. But they are fatal for a second reason which is that the dosage could cause tolerance mismatch.
Option A: This is not true. Even opioids are fatal for the same reason. When one loses tolerance, then a dosage that was comfortable earlier becomes fatal. That applies to cocaine as well as opioids. Option A is not the answer.
Option B: It is not about increasing one's tolerance or there being a limit to how tolerant you can be. Moreover, opioids are not just fatal in such cases. They are also fatal, like with cocaine, when one takes a previous dosage after losing tolerance. Option B is not the answer.
Option C: Opioids are not fatal only when a dosage causes tolerance mismatch. This is one of the scenarios. The other scenario is when one loses tolerance and then consumes a previously comfortable dosage. Option C is not the answer.
Option D: This is the difference between opioids and cocaine. Cocaine according to the passage is fatal when one loses tolerance for a previous dosage. Opioids on the other hand can be fatal even when one doesn't lose their tolerance, because of the eventual mismatch between higher-dosage-tolerant reward pathway and lower-dosage-tolerant respiratory pathway. Option D is the answer.
Choice (D)

7. Option A: The first line of the fourth para clearly mentions that eliminating gut bacteria (which is what the antibiotics are understood to be doing) negate opioid tolerance which decreases reward-seeking behaviour. So, we can infer that opioid tolerance and reward-seeking behaviour are directly proportional to each other. One goes up, the other goes up as well. Option A is not the answer.
Option B: Antibiotics allowed the mice to tolerate more pain because according to the last line, their tolerance for morphine had decreased after being administered antibiotics. So, Option B cannot be inferred, quite the contrary. Option B is the answer.
Option C: If this were not true, the rest of the para wouldn't make any sense. The author made the inferences about gut bacteria being responsible for opioid tolerance only because it was understood that antibiotics eliminate gut bacteria. Option C is not the answer.
Option D: The para clearly states that mice felt the effects of morphine more with antibiotics thereby tolerating more pain. So, morphine tolerance goes down allowing mice to feel effects of morphine more, thereby withstanding more pain. Option D can be understood. Option D is not the answer.
Choice (B)

8. Option A: This is not true, since the passage clearly mentions that antibiotics do kill gut microbiota. That is why the experiments were possible. Option A is easy to eliminate.
Option B: It is true that opioid usage changes the nature of the microbiota according to the previous para. Hence, the microorganisms present in mice exposed to morphine are different from the gut microbiota of mice not exposed to morphine, and also different from microbiota of mice going through withdrawal. The experiment however induces withdrawal in the other mice (morphine-naïve). It doesn't compare with the microbiota of mice going through withdrawal with microbiota of mice exposed to morphine. Hence, Option B is not the answer.
Option C: The experiment was simple. They put the gut microbiota of mice going through opioid withdrawal into mice that have never had morphine. The latter showed the withdrawal symptoms shown by the first. This proves that the gut microbiota has something to do with those symptoms and it is not the same as the one in mice not exposed to morphine (opioid usage changes the nature of microbiota twice as mentioned in the previous para). So, this option is apt. Option C is the answer.
Option D: The para doesn't connect withdrawal symptoms with defecation anywhere. The faeces were simply the source of the gut microbiota. Option D is not the answer.
Choice (C)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 469

9. The author is quite vehement in his statement that even the most radical and unconventional scientific ideas and breakthroughs are "always historically motivated" and are based on internal contradictions in existing theories and, also on overwhelming results from scientific experiments that forces scientists to accept such contradictions. In short, he says that science works in continuity- meaning, it is based on existing ideas. Examples that illustrate this point are not the correct answer.
Option A: This option states that Einstein's general theory of relativity, while radical, is based on existing theories of gravity and special relativity. This is a clear example of the author's point and so, it is not the answer.
Option B: While Copernicus did not entirely rely on existing ideas, his famous theory about the heliocentric system was made to address the contradictions he found in the existing theory of Ptolemaic astronomy. This illustrates the author's idea about theories being based on "well-informed analysis of the internal contradictions within existing, successful theories". Hence, Option B is also not the answer.
Option C: The option talks about the confirmation of Standard Model theory as opposed to newer radical alternatives, which indicates the discontinuity has not worked. This is also not the correct answer.
Option D: From the word "invent" we glean that the concept of calculus was entirely new and not based on existing ideas or theories. This is true for both Newton and Leibniz. Also, there is no indication in the option that this invention was based on historical scientific theories. Option D is the correct answer.
Choice (D)

10. The author is of the opinion that while Popper and Kuhn, who are popular due to their interesting theories, have an incomplete picture of science which has ended up misleading research. Then he goes on to cite that the result of this is methodological confusion. Due to their insights, physicists believe that past knowledge and historical science are unimportant and new arbitrary theories can be formed out of the blue.
Option A: The author talks about methodological confusion where theoretical work consists of "pulling arbitrary possibilities out of the blue and developing them" to further the interests of science. This is something that Popper and Kuhn would have endorsed in their theories. It is not the answer.
Option B: In the passage, we see that the author condemns the confusion caused by Popper and Kuhn which led to physicists believing that past knowledge/historical science is irrelevant to new pursuits. This option states that ideas, however random they may seem, are formed on the basis of historical scientific inventions – it is the opposite of the ideas propagated by Popper and Kuhn and is the correct answer.
Option C: The author states that this "methodological philosophy has given rise to much useless theoretical work in physics and many useless experimental investments" and that "arbitrary jumps in the unbounded space of possibilities have *never* been an effective way to do science". The philosophy here refers to Popper and Kuhn's ideas. This opinion would be something they would recommend and is not the answer.
Option D: Similar to options A and C this option endorses the ideas of Popper and Kuhn, since the author mentions that this has led to theoretical physicists believing that speculation could lead to untested ideas (with no basis in historical scientific inventions), which in turn, could lead to breakthroughs in science. Hence, Option D is not the answer.
Choice (B)

11. Point I: The author is of the opinion that there are different kinds of philosophies and the "wrong philosophy of science is held dear today by many physicists". This does not mean that the author implies that any kind of philosophy in science will cause methodological confusion. This option generalizes what is given in the passage and is incorrect.
Point II: In the passage, the author talks about sterility in theoretical physics due to physicists adopting wrong philosophies. He states the example of Popper and Kuhn and elaborates on how their ideology – due to its wide acceptance in the community of theoretical physicists – has led to useless experimentation. Since one of the reasons for the sterility is the wrong philosophy, discounting the philosophy helps in furthering the interest of science. This point is correct.
Point III: In the passage, the "why not" philosophy refers to Popper and Kuhn's philosophy which has caused methodological confusion. The author states that theoretical physicists have the misconception that this philosophy opens new doors and leads them further down the knowledge trail. The author does not agree with this and so, this cannot be considered as the author's stance.
Point IV: The author strongly states that the right philosophy in science would be to base theoretical work on the strong foundation of existing, successful theories and "building upon a vast and ever-growing *accumulation* of empirical and theoretical knowledge". Hence, this point is correct.
Choice (C)

12. The central argument of the author in the passage is that theoretical physicists are unable to produce quality research owing to the methodological disaster of arbitrary research that has no basis, as opposed to what they are supposed to do: look to verifiable historical scientific knowledge for insights. Any argument that counters this would be the ideal counterargument.
Option A: In the first line of the passage, the author mentions that the field of theoretical physics seems to be relative sterile and the cause for this is that the wrong philosophy is being followed by contemporary theoretical physicists. This is the background information that the author gives to make his main argument. This option talks about how this sterility is not unique to the said timeframe and has occurred several times in scientific history and was followed by bouts of scientific breakthroughs. This point does nothing to address the author's arguments. Option A is not the answer.
Option B: This option refutes the idea that there is a wrong methodological philosophy. Irrespective of what philosophy is applied, in the end, theories are formed from arbitrary possibilities and independent of empirical past knowledge. It counters the author's central argument and is the correct answer.
Option C: This option talks about validity of incorporating past knowledge in current research. This is in line with what the author suggests. This is not the correct answer.
Option D: This option talks about confusion in "the minds of theoretical physicists" due to the constraints in and rigidity of empirical/past knowledge, while also stating that this hindrance can be overcome when testing un-falsified arbitrary theories. This points to a problem with the mindsets of physicists rather than an issue with the ideology itself. Hence, Option D is not the answer.
Choice (B)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 561

13. Option A: The author mentions that while borrowing may not produce an optimal business model in a new-market enterprise, it does help with lowering costs in terms of "money and time needed to design a good-enough-for-now offering". Also, the author is of the opinion that striving for an optimal solution is "an unrealistic and unnecessary aim in a brand-new market". Hence, Option A is not the correct answer.
Option B: It is suggested in the passage that instead of investing on finding optimal solutions, new-market enterprises could "spend more on other aspects of the business model and on testing assumptions". However, this is not related to the possibility of borrowing faulty ideas. Hence, Option B is not the answer.
Option C: While it is true that new-market enterprises need to assemble a working prototype by borrowing ideas from peers, this does not address the question about the risk of borrowing faulty ideas. Indeed, this need to borrow ideas instead of investing in an optimal solution can be the cause of incorporating faulty ideas borrowed from peers. Option C is incorrect.
Option D: The author mentions that while it is sometimes risky to borrow ideas – they might turn out to be faulty , this risk is mitigated by new-market enterprises focussing on "how to profitably deliver value to customers" and because of this "they're likely to be reasonably astute judges of whether a given idea is good". Option D correctly captures this point and is the answer.
Choice (D)

14. Option A: We can see in the passage, the author talks about children to introduce the concept of borrowing – how children observe and learn from one another, rather than competing or coming up with entirely new ideas or patterns of behaviour. Similarly, this concept of borrowing can also be observed in new-market innovators. Option A can be inferred and is not the answer.
Option B: From the last line of the para, we understand that "astute" borrowing – borrowing the right ideas – makes all the difference between a successful new-market innovator and a failure. Option B is also incorrect.
Option C: The author mentions that young children learn from each other, by means of observation, which shows that different children have different skill sets which they learn by imitating their peers. It is obvious that when learning by themselves, they learn different things, which they can teach each other. Hence, Option C is not the answer.
Option D: In the first para, the author mentions that pre-schoolers/children rarely "bother trying to outdo one another", which mean that they do not compete; rather, they learn from each other. We cannot infer from the first para that competition coupled with borrowing helps with attaining success. Hence, Option D is the correct answer.
Choice (D)

15. In the second para of the passage, the author discusses the practice of borrowing as opposed to the conventional strategy of differentiation, when it comes to new-market enterprises. From this para, we come to understand/infer the attributes that the strategy of differentiation does or doesn't ascribe to.
Option A: Traditional strategists argue that the strategy of differentiation is important so as to avoid competition that is only cost-based. If a product is developed based on borrowed ideas, then the only way it competes with other similar products is through cost/price. The competition that occurs through implementing a strategy of differentiation will lead to competition that is not just cost-based. Option A suggests the opposite and is not the answer.
Option B: This strategy avoids cost-based competition. This is an attribute that the strategy of differentiation ascribes to and is the correct answer.
Option C: In the passage, the author mentions that "trying to differentiate early on in a new market *can lead* a company down a blind alley" but, this is not a given. The author only says that it is likely that in some cases, progress may not be possible by employing this strategy. Option C is incorrect.
Option D: The author mentions that a more effective approach, as opposed to the strategy of differentiation, is considering other companies as peers rather than as competitors. This option runs directly counter to the point that the author mentions and so, Option D is not the answer.
Choice (B)

16. The author uses fintech companies as examples to outline how the concept of borrowing is beneficial to new-market businesses. Throughout the passage, the author maintains the stance that these companies should focus on competing with existing products from established companies, rather than their new-market peers. From this, we can understand that focussing on anything else can be considered as "worrying about the wrong things".
Option A: The author is specifically of the opinion that new-market peers should be considered as a "treasure trove of ideas" rather than as competition. Thus, focussing on competing against them can be considered as worrying about the wrong thing. Option A is not the answer.
Option B: This option talks about considering existing products and established brands as competition. This is the point that the author is trying to make – that new-market enterprises should be worrying about this. We understand this from the line "their primary competitive focus should almost always be on an existing substitute" and from the example "successful fintech companies…saw their true competition as established investment and wealth-management firms". Hence, option B is the correct answer.
Option C and D: Both of these points endorse competing with new-market peers – by marketing products as superior or by competitively pricing products. These are prime examples for the phrase "worrying about the wrong things" and so, neither option is the correct answer.
Choice (B)

Solution for question 17:

17. Option A: Imagination doesn't posit the object as non-existent according to the para. It is posited as ***not existing here and now.*** It is intended as a virtual object. Option A is incorrect, and is, hence, not the answer.
Option B: While imagination is indeed a certain kind of awareness, that awareness is of a virtual object, something that is not here in the immediate vicinity. Such an object, though, may not be non-existent. That object may be for real, just not present here. Hence, Option B is easy to eliminate.
Option C: This option represents all the ideas, that both imagination and perception represent intentionality, and that imagination is the awareness (consciousness) of something that is not present here (maybe existent elsewhere), different from perception, which posits an existent object. So, the difference between imagination and perception have also been depicted correctly. Option C is the answer.
Option D: While it is true that imagination exemplifies or demonstrates the power of consciousness, the para doesn't really undermine perception or bring in a comparison between imagination and perception, as far as exemplifying the power of perception is concerned. Hence, Option D doesn't represent the essence of the para.
Choice (C)

Solution for question 18:

18. Sentence 1 is an independent sentence that defines what progressivism is.
Sentence 2 is an independent sentence that explains five parameters that define progress.
Sentence 3 refers to a previous sentence that talks about 'these ideas'.
Sentence 4 talks about the rise of Progressivism in Europe.
It can be understood that 23 is a block, because 2 introduces the ideas of progress and 3 references 'these ideas.' Similarly, it can be understood that 3 can come only after 4 since 4 introduces the Age of Enlightenment (which is needed to discuss 'figures' or leaders of enlightenment in 3). Also, 1 introduces Progressivism (so, it comes before 4). We also can see that 1 comes before 2 because even progress is introduced in 1 (as that which asserts the advancement…) and not in 2 (another reason why the 14 block should come before the 23 block). So, the order is 1423 as 23 cannot be separated.
Ans: (1423)

Solutions for questions 19 and 20:

19. Option A: The interpretations carry misconceptions, true. However, the para has another important idea – the cause behind why the interpretations carry misconceptions. It is the absence of definitive objective criteria. That part has been left out. Option A is not the answer.
Option B: While the awareness for original historical conditions as an important criterion has been mentioned in the para, the para doesn't say interpretations have to be objective. There are objective criteria to make the interpretations. The interpretations need not be objective per se. Option B is not the answer.
Option C: This option seems to indicate that innovative and persuasive interpretations reflect historical conditions/aspects. That is not always true. When it is true, they could be magnificent even when wrong. Moreover, there aren't definitive guidelines, and that is what leads to misconceptions. Hence, Option C is not the answer.
Option D: This option mentions both the big ideas of the para - that there are no definitive guidelines, and that every interpretation ends up having a misconception, probably because of the absence of guidelines. Option D is the answer.
Choice (D)

20. Option A: The historical process involves man's acquiring of a nature of his own – this part represents the conclusion of the para, in the last sentence. It is the main idea. The acquiring of a nature from the historical knowledge of human activities performed by others is the other important idea presented in the first half of the para. Option A depicts the main idea and sub-idea in the para. Option A is the answer.
Option B: Human nature is not necessarily the 'sum total' of all the operations which we have learned to perform from others. While that knowledge exists, human nature may be only a part of what we know, because we are recreating for ourselves this or that kind of human nature, not a human nature that engenders everything. Option B is not the answer.
Option C: This option can be easily eliminated because it likens human nature to historical knowledge, which is incorrect. Human nature is created by the ability of humans to recreate the past based on the historical knowledge. Human nature itself is not directly historical knowledge. Option C is not the answer.
Option D: The creation of human nature according to the last sentence of the para, happens based on the recreation of the past – the historical knowledge of human activities performed by everyone in the past - which he acquires. The last line also says 'creates for himself this or that kind of human nature'. So, it doesn't have to be a sum total of all the knowledge. The human nature is not necessarily an amalgamation of all the knowledge. Option D is not the answer.
Choice (A)

Solution for question 21:

21. The statement preceding blank (1) talks about how, if an investigation is carried out, the factor in question will be seen to function by discrete quanta with gaps between them. The statement succeeding blank (1) explains what is apparent when the above investigation is carried out. Therefore, these two statements are connected and inserting the highlighted sentence in blank (1) would break the flow. Hence, the highlighted sentence does not fit in blank (1).
The statement succeeding blank (2) talks about how the investigation mentioned prior to blank (1) can be extended. Hence the highlighted sentence should add to the previous investigation. The sentence preceding blank (2) talks about having an apparently definitive answer, but the highlighted sentence talks about everything impinging on everything, although in this case faintly. Since the preceding sentence does not talk about any kind of overlap, these two are not connected. Hence, the highlighted sentence does not fit in blank (2).
The statement preceding blank (3) talks about how further investigation would lead to a less definite answer and that the picture would have to be enlarged. The highlighted sentence fits with the idea of what would happen when the picture is enlarged. One would observe that everything impinges on everything else. The statement succeeding blank (3) does not seem to be directly linked with the statement preceding blank (3) but the "systems" mentioned in the highlighted sentence is related to the "systems" mentioned in the statement succeeding blank (3). Besides, there has been no other mention of "systems" in the paragraph. Hence, the highlighted sentence fits in blank (3).
The highlighted sentence cannot fit in blank (4) because it does not follow the idea of the existence of closed systems mentioned in the statement preceding blank (4). Nor does it summarize the paragraph. Hence, the highlighted sentence does not fit in blank (4).
Choice (C)

Solution for question 22:

22. Sentence 1 is a logically dependent sentence referencing to a previous idea and pointing out that 'that's ultimately what we need to explain'.
Sentence 2 asks us to read 'the math' carefully to realise 'something' feels like a flow of time. Since it starts with "However", it needs a precedent.
Sentence 3 discusses a theory (we shouldn't think that things will feel the way they actually are) and ends with how physics has many such examples. Also, 'while doing so" requires a precedent.
Sentence 4 talks about how the flow of time is an illusion.
2 introduces the idea of the math feeling like a flow of time. Since it starts with "However", the statement preceding this has to contrast it and should state something along the lines of "the math not feeling like a flow of something." This is what 4 talks about – "nothing flowing about math". So, 42 is a block. The flow of time is an illusion…in spacetime. If you study it…feel like a flow of time. Similarly, we can notice that 13 is a block. 1 tells us what we need to explain, and 3 tells us we shouldn't be so naive as to think the other way around. 'While doing so" in 3 refers to "trying to explain Why we feel…" in 1. So, even though it is a pair, 3 cannot come first. 1 needs to follow 2 – 'that's ultimately what we need to explain' references a previous sentence and then asks 'why does everything feel the way it does'. So, the previous sentence has to talk about something feeling like something else – which has been established in 42. So, 4213.
Another possibility is that the order could be 1342. But there is no direct connection between 4 and 3 as they seem to talk about different ideas. Therefore, only 4213 forms a coherent paragraph.
Ans: (4213)

Solution for question 23:

23. The statements immediately preceding and following blank (1) talk about how gossip and information is exchanged at the office watercooler. These statements talk about gossip specific to the workplace. The highlighted sentence talks about gossip and how it is useful at an individual level. Hence, the highlighted sentence does not fit in blank (1).
The statement preceding blank (2) talks about the kind of gossip shared at the workplace and the statements following blank (2) question our concern for gossip. The highlighted sentence talks about how intelligent gossip is good. Neither of the said statements discuss intelligent gossip. Hence, the highlighted sentence does not fit in blank (2).
The statement preceding blank (3) states that we can benefit from the informed opinion of others. The statement following blank (3) adds to this by stating that we anticipate how colleagues and friends will evaluate our choices. The highlighted sentence combines these two statements together and states that the expectation of intelligent gossip is a powerful motive for self-criticism. Therefore, the highlighted sentence has to follow these two statements and cannot come in between them. Hence, the highlighted sentence does not fit in blank (3).
The statements preceding blank (4) talk about how we can benefit from gossip and that we anticipate how our friends and colleagues evaluate our choices. "the expectation" mentioned in the highlighted sentence refers to the anticipation mentioned in the statement preceding blank (4). Because the quality and the content of the anticipated judgements matter, the expectation of intelligent gossip is a powerful motive for serious self-criticism. This is difficult otherwise, as stated in the statements between blanks (2) and (3). Hence, the highlighted sentence fits in blank (4).
Choice (D)

Solution for question 24:

24. Sentence 1 references another sentence with 'these' resources.
Sentence 2 is also a dependent sentence referring to 'their' administration.
Sentence 3 is an independent sentence that gives us what the resources spoken about in 1 are – soil, water, etc.
Sentence 4 talks about the utilisation and preservation of something. We can infer that it is the resources whose utility for present needs and preservation for future generations is being discussed.
It can be understood that 3 establishes background and context. So, it has to be the starting sentence of the para, more so because all the other sentences seem to depend on the subject of resources. All the other three sentences are talking about the resources in 3 (which is precisely what should stop you from putting 1 after 3, because even 2 and 4 have pronouns referring to the resources). However, 4 is the most upstream of all the three. It talks about the importance of these resources which is a bigger idea compared to who is responsible for these resources and who isn't. 4 also talks about the nature of the program to preserve the resources.
Between 1 and 2, 2 has to come first because 2 is a negative sentence, with a sense of incompleteness. The para cannot end at an opinion that 'the administration' is not a matter of politics (the obvious question would be 'then what?'). It makes more sense for 2 to come first followed by 1, which continues the idea of 2 – cannot be a matter of politics – by long tradition has been taken care of by men of professional statures. So, 3421 is the order.
Ans: (3421)

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	2, 6, 15, 16
Difficult	1, 3, 4, 5, 7, 8, 9, 10, 13, 14, 23, 24
Very Difficult	11, 12, 17, 18, 19, 20, 21, 22

SECTION – II

Solutions for questions 1 to 5:

Eight persons are from three countries, play three sports and work in three countries. From the given description, we can see that three persons belong to each of two countries and two persons belong to the third country. Same is the case for the sports that they play and the companies that they work in.

From (i), the averages given are 190, 125 and 160. Of these, two averages are average of three numbers, while one is an average of two numbers.

If 125 is an average of three numbers, the sum of these three numbers will be 375, which is not possible as all the salaries are multiples of 10. Hence, 125 is an average of two numbers, while the other two are averages of three numbers.

Hence, three persons play Tennis, three persons play Football and two persons play Cricket.

Similarly, from (ii), three persons are from India, three persons are from China and two persons are from Sri Lanka. And from (iii), three persons work in HP, three persons work in Microsoft and two persons work in Dell.

The following tables provide the sum of the salaries of the eight persons, according to the country they are from, the company that they work in and the sport that they play:

Country	No.	Total Salary
India	3	660
China	3	510
Sri Lanka	2	130

Company	No.	Total Salary
HP	3	480
Microsoft	3	510
Dell	2	310

Sport	No.	Total Salary
Tennis	3	570
Football	3	480
Cricket	2	250

We also know six of the eight salaries: 230, 320, 110, 50, 140 and 200.
Also, the total salary of the eight persons = 3 × 660 + 3 × 510 + 2 ×130 = 1300.
Hence, the salaries of B and E must add up to 1300 – 230 – 320 – 110 – 50 – 140 – 200 = 250.
We need to distribute these numbers, along with another two (the salaries of B and E), into the three buckets for each category.
We can start with Sri Lanka, as it is the lowest of all the values calculated above (values which are at either extremes usually will result in lower number of possibilities).
The sum of two salaries is 130. From the above values, we can get two possibilities – (110, 20), where 20 is the salary of either B or E AND (50, 80), where 80 is the salary of either B or E.
However, if 20 is the salary of either B or E, then the other person must have a salary of 230. This is not possible as all the salaries must be distinct and A has a salary of 230.
Hence, one of B and E must have a salary of 80 and the other must have a salary of 170. From (iv), B must have a salary of 170 and E must have a salary of 80.
For Sri Lanka, the two salaries must be 80 (E) and 50.
For India, the sum of salaries is 660. This is possible only if C is from India. Otherwise, the sum of the next three highest salaries will not add up to 660. The other two salaries must add up to 660 – 320 = 340. The other two persons who can be from India are (A, D) OR (H, G). The salaries of these two pairs are (230, 110) and (200, 140). In both the cases, B must be from China. The other two persons from China must also have a salary of 340. The only possible pairs are (A, D) and (H, G).
From (iv), B plays Tennis. Since the sum of the salaries of the persons who play Tennis is 570, the sum of the salaries of the other two persons must be 570 – 170 = 400.
This is possible only if C and E, with salaries of 320 and 80, play Tennis.
Also, the sum of the salaries of the persons who play Cricket is 250. From this, we can see that A cannot play Cricket (since his salary is 230 and there is no other person with a salary of 20). Hence, A plays Football.
From (iv), B works in Dell. Hence, the salary of the other person who works in Dell is 310 – 170 = 140. Hence, G also works in Dell.
The sum of the salaries of the persons working in Microsoft is 510. The only possibilities for this are (320, 110, 80) and (230, 200, 80). In either case, E from Sri Lanka must work in Microsoft.
Also, in either case, F cannot work in Microsoft. Since F cannot work in Dell as well, F must work in HP.
From (v), two persons from the same country work in Microsoft. E cannot be one of the two persons because F, who is from the same country as E, cannot work in Microsoft.
If the salaries of the persons working in Microsoft are (320, 110, 80), C and D must belong to the same country. This is possible.
If the salaries of the persons working in Microsoft are (230, 200, 80), A and H must belong to the same country. However, this is not possible as we have already established that between (A, D) and (H, G), one pair belongs to India and one belongs to China. Hence, A and H cannot belong to the same country. ∴ This case is not possible.
The salaries of the person working in Microsoft must be (320, 110, 80). C and D work for Microsoft and must belong to the same country. Since C belongs to India, D must also be from India. A and D must be from India and H and G must be from China.
From (v), the two persons who play Cricket work in the same company. And the sum of their salaries must be 250. This is possible only if H and F play Cricket.
D and G must play Football.
The following table provides the distribution:

Person	Salary (in USD '000)	Country	Sport	Company
A	230	India	Football	HP
B	170	China	Tennis	Dell
C	320	India	Tennis	Microsoft
D	110	India	Football	Microsoft
E	80	Sri Lanka	Tennis	Microsoft
F	50	Sri Lanka	Cricket	HP
G	140	China	Football	Dell
H	200	China	Cricket	HP

1. The salary (in USD '000) of B is 170. Ans: (170)
2. F works in HP. Choice (C)
3. A and D are both from India and play Football. The sum of their salaries is 340. Ans: (340)
4. For two countries, both Sri Lanka and China, it can be said that each person from that country plays a different sport. Choice (B)
5. D works in Microsoft and plays Football. Choice (B)

Solutions for questions 6 to 10:

From the graph, we can find the number of students who scored more than Ravi and less than him in each of the ten tests. This is provided in the following table:

Test	No. of students who scored less	No. of students who scored more
Test 1	20	179
Test 2	280	519
Test 3	300	99
Test 4	455	244
Test 5	300	299
Test 6	360	539
Test 7	540	59
Test 8	560	239
Test 9	135	164
Test 10	630	269

6. The minimum number of students who scored better than Ravi in any of the ten tests is 59. Ans: (59)
7. Ravi's rank was less than 250 in six tests – Test 1, Test 3, Test 4, Test 7, Test 8 and Test 9. Choice (D)
8. The ten tests as ranked by Ravi would in the order Test 7, Test 3, Test 9, Test 1, Test 8, Test 4, Test 10, Test 5, Test 2 and Test 6. Hence, Test 1 would have been ranked fourth. Choice (C)
9. Test 4 would have been ranked sixth. In this test, 244 students scored more than Ravi. Ans: (244)
10. In Test 2, 519 students scored more than Ravi and 280 students scored less than him. This test was attempted by 800 students.
In Test 6, 539 students scored more than Ravi and 360 scored less than him. This test was attempted by 900 students.
Of the 519 students who scored more than Ravi in Test 2, 360 students could have scored less than him in Test 6 (since the number of students who scored less than him was 360 in Test 6).
The remaining 519 – 360 = 159 students must have scored more than Ravi in both the tests.
Hence, a minimum of 159 students would have scored more than him in both Test 2 and Test 6. Choice (D)

Solutions for questions 11 to 15:

The five cities are ranked from 1 to 5 in three parameters. The sum of the ranks of all the five cities across the three parameters is 45.
Since the sum of the ranks of each city was the same, the sum of the ranks of each city must be 45/5 = 9.
The possible ways in which the sum of three ranks can add up to 9 are (1, 4, 5), (2, 2, 5), (2, 3, 4), (3, 1, 5), (3, 3, 3) and (4, 1, 4).
From (iii), Ahmedabad was not ranked last in any parameter. Hence, the possible ranks for Ahmedabad are (2, 3, 4), (3, 3, 3) and (4, 1, 4). From (iii), only Kolkata and Indore received the same rank in two parameters. Hence, Ahmedabad could not have received ranks of (4, 1, 4). Only two possibilities exist for the ranks of Ahmedabad – (2, 3, 4) and (3, 3, 3).
From (iv), the rank of Ahmedabad in I is the same as that of Bangalore in C, and from (ii), this is the same as that of Lucknow in S.
If Ahmedabad received ranks of (3, 3, 3) then no other city could have received the third rank in any parameter. However, in C, Bangalore must have received the same rank as Ahmedabad in I. Hence, Ahmedabad must have received ranks of (2, 3, 4).

From (iii), Indore could have received ranks of (2, 2, 5) or (4, 1, 4). If Indore received (2, 2, 5), then in I, Indore must have received 5th rank or 2nd rank. However, from (ii), Indore was not ranked last in I. If Indore received 2nd rank in I, then, from (i), Ahmedabad must have received 1st rank in I, which is not possible. Hence, Indore could not have received (2, 2, 5) and must have received (4, 1, 4).
From (i), Indore must have received 4th rank in I.
Since Indore received two fourth ranks and Ahmedabad received one fourth rank, no other city could have received fourth rank in any parameter. Further, Ahmedabad could not have received fourth rank in I, since both Bangalore in C and Lucknow in S have the same rank. Hence, all these three ranks can only be 2 or 3.
Hence, from (iii), Kolkata must have received ranks of (2, 2, 5). From (i), Kolkata received 2nd rank in S.
Since Kolkata received two second ranks and Ahmedabad received one second rank, no other city could have received a second rank. Hence, both Bangalore and Lucknow must have received (1, 3, 5) ranks.
Since Bangalore could not have received 2nd rank in any parameter, Bangalore in C, Ahmedabad in I and Lucknow in S must have received 3rd rank.
In I, Bangalore and Lucknow must have received 1st and 5th rank in any order. Hence, Kolkata must have received 2nd rank in I. From this, it follows that Kolkata received 5th rank in C.
Since Kolkata received 5th rank in C, Lucknow must have received 1st rank in C; Indore must have received 4th rank; and Ahmedabad must have received 2nd rank.
Since Lucknow received 1st rank in C, it must have received 5th rank in I; Bangalore must have received 1st rank in I and 5th rank in S.
Ahmedabad must have received 4th rank in S, while Indore must have received 1st rank in S.

The following table provides the ranks of the cities in the three parameters:

City	Cleanliness	Infrastructure	Safety
Ahmedabad	2	3	4
Bangalore	3	1	5
Kolkata	5	2	2
Indore	4	4	1
Lucknow	1	5	3

11. Ahmedabad was ranked 4th in Safety. Choice (A)
12. Two cities (Kolkata and Indore) were ranked better than Lucknow in both Infrastructure and Safety. Choice (C)
13. The rank of Kolkata in Cleanliness is the same as the rank of Bangalore in Safety. Choice (C)
14. Indore was ranked first in Safety. Choice (D)
15. Since the sum of the ranks is the same for each city and no two cities have the same rank in any parameter, for any pair of cities, one city cannot be ranked better (or worse) in each of the three parameters than the other. Hence, for any pair of cities, one of the cities will be ranked better than the other in at least two parameters. For example, Ahmedabad is ranked better than Bangalore in two parameters and Ahmedabad is ranked worse than Kolkata in two parameters. Hence, we should count both the pairs, since one city is ranked better than the other in at least two parameters. Therefore, the given condition is satisfied for all possible pairs of cities, i.e., for 4 + 3 + 2 + 1 = 10 pairs of cities.

Note that to answer this question, we need not solve the entire set. Understanding the implications of the sum of ranks being same for all the cities will be sufficient to answer this question.
Choice (A)

Solutions for questions 16 to 20:

Given that no team won all the matches in the tournament. Also, from (ii), D lost two matches, against A and B, and won the match against C.

From the table, we can see that the total points scored by D is the sum of the goals scored by and against D. This is possible in only the following cases:
(a) D won all the matches – in this case, the total points will be sum of the goals scored by and against D.
(b) D lost one or more matches but did not score any goals in the matches that it lost: in this case, the number of points awarded to D even in the match that it lost will be the total number of goals scored in the match (since in this case, the difference between the number of goals scored by the two teams is the same as the total number of goals scored by the two teams).

Since it is given that D lost two matches, in the two matches that it lost, D must not have scored any goals. In the match that D won, D must have scored all the 4 goals that it scored in the tournament. From (ii), this match must be against C.

From (ii), A scored 4 goals in the match against D. Hence, in the match between A and D, A must have scored 4 goals and D must have scored 0 goals.
Since A scored 4 goals against D, in the matches against B and C, A must have scored 6 goals, in total (as A scored a total of 10 goals). B and C must have scored 5 goals against A in these two matches. Also, A scored 4 points in the match against D. Hence, A must have scored 3 points in the matches against B and C.
The minimum number of points awarded to any team for a match is 1 (since the loser gets the difference of points, which can be 1 or the winner can get 1 point in which only one goal is scored).
Hence, in the two matches, A must have scored 1 point and 2 points in some order. Also, since A won against D, from (i), A should have won one of these two matches and lost the other.
If A won the match in which it scored 1 point, then the final score of the match must be 1 – 0. Hence, in the other match A must have scored 5 goals (since A scored 6 goals in these two matches) and the opposite team must have scored 7 goals (so that A gets 2 points from this match). In this case, the total number of goals scored against A in the two matches is 0 + 7 = 7. However, only 5 goals were scored against A in the tournament. Hence, this case is not possible.
Hence, A won the match in which it scored 2 points and lost the match in which it scored 1 point. To win a match and score only 2 points, the only possibility is if A scored 2 goals and the opposite team scored 0 goals. Hence, in the match that A lost, A must have scored 4 goals (since it scored 6 goals in these two matches) and the opposite team must have scored 5 goals. Since the total number of goals scored against B is only 3, the match in which A scored 4 goals must be against C (and C scored 5 goals in this match).
Against B, A scored 2 goals and D did not score any goals. Hence, C should score 1 goal against B. Against C, D scored 4 goals and A scored 4 goals. Hence, B should score 3 goals. Hence, in the match between B and C, B scored 3 goals and C scored 1 goal.
Since B scored 3 goals against C and 0 goals against A, B must score 3 goals against D (as B scored a total of 6 goals in the tournament).
Since C scored 1 goal against B and 5 goals against A, C must score 1 goal against D (as C scored a total of 7 goals in the tournament).

The following table provides, for each match, the goals scored by and against each team and the points scored by the winning team and the losing team (the winning team and losing team are indicated in parenthesis alongside the points):

Match	Goals By	Goals By	Points to Winner	Points to Loser
A – B	A: 2	B: 0	2 (A)	2 (B)
A – C	A: 4	C: 5	9 (C)	1 (A)
A – D	A: 4	D: 0	4 (A)	4 (D)
B – C	B: 3	C: 1	4 (B)	2 (C)
B – D	B: 3	D: 0	3 (B)	3 (D)
C – D	D: 4	C: 1	5 (D)	3 (C)

16. C scored 5 goals against A. Choice (C)
17. In the match between B and C, B was awarded 4 points, while C was awarded 2 points. Hence, a total of 6 points were awarded. Ans: (6)
18. B scored 3 goals against D. Ans: (3)
19. A was awarded 4 points in the match that it played against D. Choice (B)
20. In the match between A and C, a total of 9 goals were scored. This is the highest number of goals scored in a match. Choice (D)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	6, 7, 8, 9
Medium	10, 11, 12, 13, 14, 15
Difficult	1, 2, 3, 4, 5, 16, 17, 18, 19, 20
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1. Any number of the form $2^2(3^x5^y \ldots n^z)$ will have the sum of its odd factors as $(3^0 + 3^1 + \ldots 3^x)(5^0 + 5^1 + \ldots 5^y) \ldots (n^0 + n^1 + \ldots n^z)$, where n is an odd number.

The sum of its even factors will be $(2^1 + 2^2)(3^0 + 3^1 + \ldots 3^x)(5^0 + 5^1 + \ldots 5^y) \ldots (n^0 + n^1 + \ldots n^z)$, i.e., 6 times (sum of its odd factors)

Therefore, the number is of the form "4 × (odd number)". The remainder when such a number is divided by 8 is 4.

Alternative Solution:

It can be inferred from the question that the number is even. Hence, the remainder when an even number is divided by 8 can only be one of 0, 2, 4 and 6. Now, these remainders themselves can be taken, one at a time, to be the given number and checked if the other condition that the sum of all the odd factors is one-sixth of the sum of all the even factors) is satisfied.
If the remainder is 0, the number can be 8. The only odd factor is 1 and the even factors are 2, 4, 8. The condition is not satisfied.
If the number is 2, the only odd factor is 1 and even factor is 2. The condition is not satisfied.
If the number is 4, the only odd factor is 1 and the even factors are 2, 4. The condition is satisfied.
If the number is 6, the odd factors are 1, 3 and even factors are 2, 6. The condition is not satisfied.
Hence, the number can be 4, i.e., the remainder when the number is divided by 8 must be 4.
Ans: (4)

2. Let $E = \frac{x^2+6x+7}{x^2+6x+16} = \frac{(x^2+6x+16)-9}{x^2+6x+16}$

$= 1 - \frac{9}{x^2+6x+16}$

$\therefore E = 1 - \frac{9}{(x+3)^2+7}$

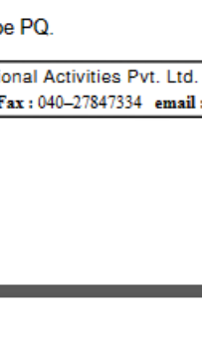
For E to be minimum $\frac{9}{(x+3)^2+7}$ must be maximum, which is the case when $(x + 3)^2 + 7$ is minimum.

$\therefore$ When x = –3

$E = 1 - \frac{9}{0+7} = 1 - \frac{9}{7} = \frac{-2}{7}$

$\therefore L = \frac{-2}{7}$ and 126L = –36 Ans: (–36)

3.



Let the pole be PQ.

The angle of elevation (θ) of P, the top of the pole is 30° at A and C and 45° at B. If the point on AB, where θ is 45° is anywhere other than B, the point directly across the road from Q, there would be two instances when the angle is 45°. As there is only one, that point (where θ = 45°) has to be B.
Let PQ = h be the height of the pole
As ∠PBQ = 45°, BQ = h
As ∠A = ∠C = 30°
$AQ = CQ = h\sqrt{3}$ m
In ΔABQ, $AB^2 = AQ^2 – BQ^2 = 3h^2 – h^2 = 2h^2$
$\therefore AB = \sqrt{2}h = 10\sqrt{2}$ m ⇒ h = 10 m Ans: (10)

4. The following are the two cases in which the composition remains unaltered after transferring one ball from A to B and then from B to A.
(i) Drawing a red ball from A and placing it in B and then drawing a red ball from B and placing it in A.
The probability of this event is
$\left(\frac{4}{4+3}\right)\left(\frac{4}{4+4}\right) = \left(\frac{4}{7}\right)\left(\frac{4}{8}\right) = \frac{16}{56}$
(ii) Drawing a black ball from A and placing it in B and then drawing a black ball from B and placing it in A. The probability of this event is
$\left(\frac{3}{3+4}\right)\left(\frac{5}{5+3}\right) = \left(\frac{3}{7}\right)\left(\frac{5}{8}\right) = \frac{15}{56}$
∴ Total probability is $\frac{16}{56} + \frac{15}{56} = \frac{31}{56}$
Choice (B)

5. If the three equations are named as (1), (2) and (3) respectively, then (1) × 2 + (2) × 3 = (3)
Hence, if P = 21, the system of equations becomes dependent and then they will not have a unique solution. For every other real value of P, a unique solution will exist. Further there is no value of P for which the equations become inconsistent. Hence statements II and III are true, while only statement I is not true. Choice (C)

6. Let $\sqrt{28-16\sqrt{3}} = \sqrt{a} - \sqrt{b}$ and hence

$\sqrt{28+16\sqrt{3}} = \sqrt{a} + \sqrt{b}$

$\therefore \left(\sqrt{28-16\sqrt{3}}\right)^2 = \left(\sqrt{a} - \sqrt{b}\right)^2$

$28 - 16\sqrt{3} = a + b - 2\sqrt{ab}$

$\therefore a + b = 28 \rightarrow (1)$

$\sqrt{ab} = 8\sqrt{3} = \sqrt{192}$

$\Rightarrow ab = 192 \rightarrow (2)$

Substituting (2) in (1) $\Rightarrow a + \frac{192}{a} = 28$

$\Rightarrow a^2 – 28a + 192 = 0$
$\Rightarrow a^2 – 16a – 12a + 192 = 0$
$\Rightarrow (a – 16)(a – 12) = 0$
$\Rightarrow a = 16$ or 12
From (1),
When a = 16, b = 12 and

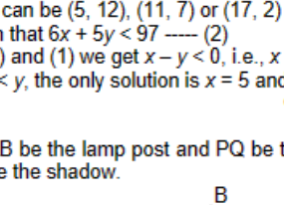
When a = 12, b = 16
∴ For $\sqrt{a} - \sqrt{b}$ to be positive a = 16 and b = 12
$\sqrt{28-16\sqrt{3}} + \sqrt{28+16\sqrt{3}} = \sqrt{a} - \sqrt{b} + \sqrt{a} + \sqrt{b}$
$= 2\sqrt{a} = 2\sqrt{16} = 8$ Choice (A)

7. If a zero is appended to the extreme right of an octal number, the number gets multiplied by 8
∴ If N was the original number
The present value of the number is $\frac{N \times 8}{25}$
Therefore, the percentage reduction in the value of the number is $\frac{17}{25} \times 100 = 68\%$ Choice (C)

8. The total simple interest on ₹60000 at 7.5% rate of interest in n years is given by $\frac{60000 \times 7.5 \times n}{100}$
Now, $\frac{60000 \times 7.5 \times n}{100} = 36000$
⇒ n = 8. Choice (D)

9. Let the number of sickles and axes that Raju purchased be x and y respectively.
Total expenditure = ₹(150x + 180y)
If he bought y sickles and x axes, his total expenditure = ₹(150y + 180x)
150y + 180x < 150x + 180y
x < y
Also, 150x + 180y = 2910
5x + 6y = 97 ------ (1)
(x, y) can be (5, 12), (11, 7) or (17, 2).
Given that 6x + 5y < 97 ----- (2)
By (2) and (1) we get x – y < 0, i.e., x < y.
As x < y, the only solution is x = 5 and y = 12.
Choice (B)

10. Let AB be the lamp post and PQ be the man. Let CP be the shadow.



Let x be the distance between man and the lamppost and y be the length of the shadow PC.
ΔABC and ΔPQC are similar. AB is $\frac{6}{1.8}$ or $\frac{10}{3}$ times PQ.
∴ x + y is $\frac{10}{3}$ times y or x is $\frac{7}{3}$ times y, irrespective of what x is (i.e. even if x continuously changes)
⇒ rate at which CP changes = $\frac{3}{7}$ × rate at which x increases.
⇒ The rate at which the length of the shadow increases
= $\frac{3}{7}$ × speed of man = $\frac{3}{7}$ × 14 = 6 m/min

Alternative Solution:

Since the man's initial position is not specified, we can take him to initially be at the bottom of the lamp post, in which case, the length of his shadow would be zero. Now, let the man walk for one minute, i.e., 14 m away from the lamp post. Now, the length of his shadow can be shown to be 6 m (using the similar triangles approach, as given above). Hence, in one minute the length of the shadow increased by 6 m. That is, the rate of increase is 6 m/min. Ans: (6)

11. Let us suppose that x men can do the work in y days. It is given that for the first 16 days all the x men worked. After this only $\frac{2}{3}$x men worked for y days and thus the work is completed
$\Rightarrow \therefore \frac{16}{y} + \frac{y}{\frac{3}{2}y} = 1$
∴ y = 48 days
Now if one-fourth of the group left after 21 days, the remaining group is 3/4 of the original. Hence remaining time taken = 4/3 × (48 – 21) = 36 days.
Hence total time = 21 + 36 = 57 days.

Alternative solution

$xy = 16x + \frac{2}{3}x{\cdot}y$
⇒ y = 48
$xy = 21x + \frac{3}{4}x{\cdot}y'$
⇒ y' = 36
∴ Total time = 21 + 36 = 57 days. Ans: (57)

12. Let Amar's speeds for the three legs OP, PQ and QR be $2V_1$, $3V_2$ and $4V_3$ -------- (1) respectively.
Given, the speeds of Amar for these three legs were in the ratio of 2 : 3 : 4.
⇒ $V_1 = V_2 = V_3 = V = 1$ say --------- (2) and from (1) the speeds of the three runners for the three legs will be

	OP	PQ	QR
Amar	$2V_1$	$3V_2$	$4V_3$
Akbar	$3V_1$	$4V_2$	$2V_3$
Antony	$4V_1$	$2V_2$	$3V_3$

Since $V_1 = V_2 = V_3 = 1$, from (2), and by assuming OP = x, PQ = 2x and QR = 3x, and x = 1.

We get times taken by Amar, Akbar and Antony as $\left(\frac{1}{2}+\frac{2}{3}+\frac{3}{4}\right)$, $\left(\frac{1}{3}+\frac{2}{4}+\frac{3}{2}\right)$ and $\left(\frac{1}{4}+\frac{2}{2}+\frac{3}{3}\right)$
respectively i.e. $\frac{23}{12}$, $\frac{28}{12}$ and $\frac{27}{12}$
∴ Amar is the fastest. Choice (A)

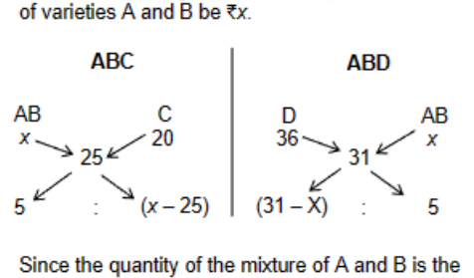
13. It is given that,
$\frac{MP-SP}{MP} \times 100 = \frac{SP-CP}{CP} \times 100$
CP (MP – SP) = MP (SP – CP)
2CP × MP = SP × MP + SP × CP
$\therefore SP = \frac{2CP \times MP}{CP+MP} = \frac{2}{\frac{1}{CP}+\frac{1}{MP}}$
∴ SP is the harmonic mean of CP and MP

Alternative Solution:

Let the CP, SP and MP be 100, 120 and 150. Clearly choices (A) and (B) are not correct checking for the HM of 100 and 150, we see that option (C) is correct.
Choice (C)

14. $S = 3 + 7x + 12x^2 + 18x^3 + 25x^4 + \text{------} (1)$
$Sx = 3x + 7x^2 + 12x^3 + 18x^4 + \text{----} (2)$
(1) – (2) ⇒
$S(1 – x) = 3 + 4x + 5x^2 + 6x^3 + 7x^4 + \text{----}$
$= 3(1 + x + x^2 + x^3 + x^4 + \text{---}) + (x + 2x^2 + 3x^3 + 4x^4 \ldots\ldots)$
$= 3\left[\frac{1}{1-x}\right] + x[1 + 2x + 3x^2 + 4x^3 + \text{----})$
$= \frac{3}{1-x} + x(1 – x)^{-2} \Rightarrow S = \frac{3}{(1-x)^2} + \frac{x}{(1-x)^3}$
$= \frac{3-3x+x}{(1-x)^3} = \frac{3-2x}{(1-x)^3}$ Choice (B)

15. Let the cost per kg of the mixture (AB) of the rice of varieties A and B be ₹x.



Since the quantity of the mixture of A and B is the same in both (ABC) and (ABD), and the quantity of D is thrice that of C, $\frac{(31-x)}{5} = 3\left(\frac{x-25}{5}\right)$.
⇒ x = 26.5
∴ The ratio of quantities of (AB), C and D is 5 : (26.5 – 25) : (31 – 26.5) i.e., 10 : 3 : 9.
Hence the cost per kg of the mixture of ABCD (in ₹)
$= \frac{(26.5)10+(20)3+(36)9}{10+3+9} = \frac{649}{22} = 29.5$
Choice (D)

16. Let us denote the number of persons standing in front of the four counters by a, b, c and d respectively.
Now a + b + c + d = 5 where a, b, c, d ≥ 0.
[Since there are a total of five persons standing in front of the four counters].
The total number of non – negative integral solutions to the above equation is ${}^{5+4-1}C_{4-1}$
$= {}^{8}C_{3} = 56$
Again, the five persons can be arranged among themselves in 5! ways.
Therefore the total number of ways = 56 × 5! = 6720 ways.

Alternative solution:

The first person can stand in front of any of the four counters i.e in 4 ways
Now, wherever the first person stands, the second person has five places (options) available and so he can stand in 5 ways.

Counter 1	Counter 2	Counter 3	Counter 4
✓	✓	✓	✓
1st person			
✓			

[We are denoting the number of positions available to the second person by the tick marks]
So, the first two persons can stand in (4) (5) ways.
Similarly, the third person can (always, i.e., irrespective of where the 2nd person stood) stand in 6 ways.

Counter 1	Counter 2	Counter 3	Counter 4
✓	✓	✓	✓
1st	2nd		
✓	✓		

OR

Counter 1	Counter 2	Counter 3	Counter 4
✓	✓	✓	✓
1st			
✓			
2nd			
✓			

Proceeding similarly, the fourth person can stand in 7 ways and the fifth in 8 ways.
Therefore the total number of ways = (4) (5) (6) (7) (8) = 6720 ways. Ans: (6720)

17. One side of the rhombus lies on the line ℓ : y = x – 1
The centre (point where the diagonals intersect each other) is the origin.
∴ The other side lies on the line which is as far from (0, 0) as ℓ.
This line has also got to be parallel to ℓ.
∴ The line has to be m: y = x + 1. Choice (C)

18. The product of the roots of $x^2 – (4 – \alpha)x + (\alpha + 1/\alpha) = 0$ is $(\alpha + 1/\alpha)$
As 2 is one root, considering the 4 options, the product of the roots would be 10/3, –2, 1 and –4.
As the product is α + 1/α, where α is some positive real number, it has to be at least 2. Only choice (A) is possible. Choice (A)

19. $\text{Rem}\left(\frac{8^{82}}{17}\right) = \text{Rem}\left(\frac{-4^{41}}{17}\right)$, since $\text{Rem}\left(\frac{8^2}{17}\right) = -4$

$= \text{Rem}\left(\frac{(-4^2)^{41}}{17}\right) = \text{Rem}\left(\frac{16^{41}}{17}\right) = -1$

Hence, required remainder = 16

Alternative Solution:

$\text{Rem}\left(\frac{8^{82}}{17}\right) = \text{Rem}\left(\frac{8^2 \cdot 8^{80}}{17}\right)$

$= \text{Rem}\left(\frac{8^2}{17}\right) \times \text{Rem}\left(\frac{8^{80}}{17}\right) = 16 \times 1$

$\left[\because \text{Rem}\left(\frac{8^{80}}{17}\right) = \text{Rem}\left(\frac{8^{(17-1)5}}{17}\right) = 1\right]$

By Fermat's theorem $\text{Rem}\left(\frac{a^{(p-1)k}}{p}\right) = 1$, where p is a prime number and a is not a multiple of p.
So, the required remainder is 16. Ans: (16)

20. $\underline{1}\,\underline{2}\,\underline{3}\,\underline{4}\,\underline{5}\,\underline{6}\,\underline{7}$
The first 3 positions can be filled in 2 ways.
Since the number is even, position seven can be filled with 0, 2, 4, 6 or 8 (in 5 ways.)
Since, the digit 7 has to appear only once, we consider the following cases.
Case (i): If the first three digits are 242 then one out of the positions 4th, 5th and 6th should be 7 and the remaining two positions can be filled in (9) (9) ways (i.e. with the digits except 7).
∴ Number of trials in this case = 3 (9) (9) (5)
Case (ii): If the first three digits are 472, then the 4th, 5th and 6th positions can be filled in (9) (9) (9) ways (i.e., with the digits except 7).
∴ Number of trials in this case = (9) (9) (9) (5)
Hence, the required number of trials = (12) (9) (9) (5) = 4860 Ans: (4860)

21. Number of girls who failed in the exam = $\frac{30}{100} \times 80 = 24$.
Number of boys who failed in the exam = 80 – 24 = 56
Total number of boys in the class = $56 \times \frac{100}{35} = 160$
Total number of girls in the class = $\frac{60}{100} \times 160 = 96$
Number of girls who passed in the exam = 96 – 24 = 72
Percentage of girls who passed the exam = $\frac{72}{96} \times 100$ = 75%. Choice (D)

22. The number of triangles of bases of different lengths are listed below

Base length	Number of triangles	Base length	Number of triangles
1	10	6	5
2	9	7	4
3	8	8	3
4	7	9	2
5	6	10	1

∴ Total area A
$= \frac{5}{2}[(1(10) + 2(9) + 3(8) + \ldots + 10(1)] + \frac{1}{2}(10)(5)$
(which is the area of triangle NOP)
Let S = 1(10) + 2(9) + ...+ 10(1)
and T = 1(1) + 2(2) + ... + 10(10) = 10(11)(21)/6 = 35(11)
∴ S + T = (1 + 2 + ...+10) (11) = 55(11)
∴ S = 20(11) and $A = \frac{5}{2}(20)(11) + \frac{1}{2}(10)(5)$
$= \frac{1150}{2} = 575$ Choice (A)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	–
Easy	6, 8, 21
Medium	4, 7, 9, 10, 11, 12, 13, 15, 17, 18, 20
Difficult	1, 2, 3, 5, 14, 19, 22
Very Difficult	16

Done
Tip: Scroll slowly for the best results